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IKEDA(10) **Pub. No.: US 2025/0277604 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **JOULE-THOMSON CRYOCOOLER**(30) **Foreign Application Priority Data**(71) Applicant: **Sumitomo Heavy Industries, Ltd.**,
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CPC **F25B 9/02** (2013.01)(21) Appl. No.: **19/211,262**(57) **ABSTRACT**(22) Filed: **May 18, 2025****Related U.S. Application Data**(63) Continuation of application No. PCT/JP2023/
038530, filed on Oct. 25, 2023.

A Joule-Thomson cryocooler includes a pre-cooling cryo-cooler that includes a pre-cooling stage, a stage extension component that is attached to the pre-cooling stage and is cooled by the pre-cooling stage, and a refrigerant pipe that is mounted on the stage extension component to enable heat exchange with the stage extension component.

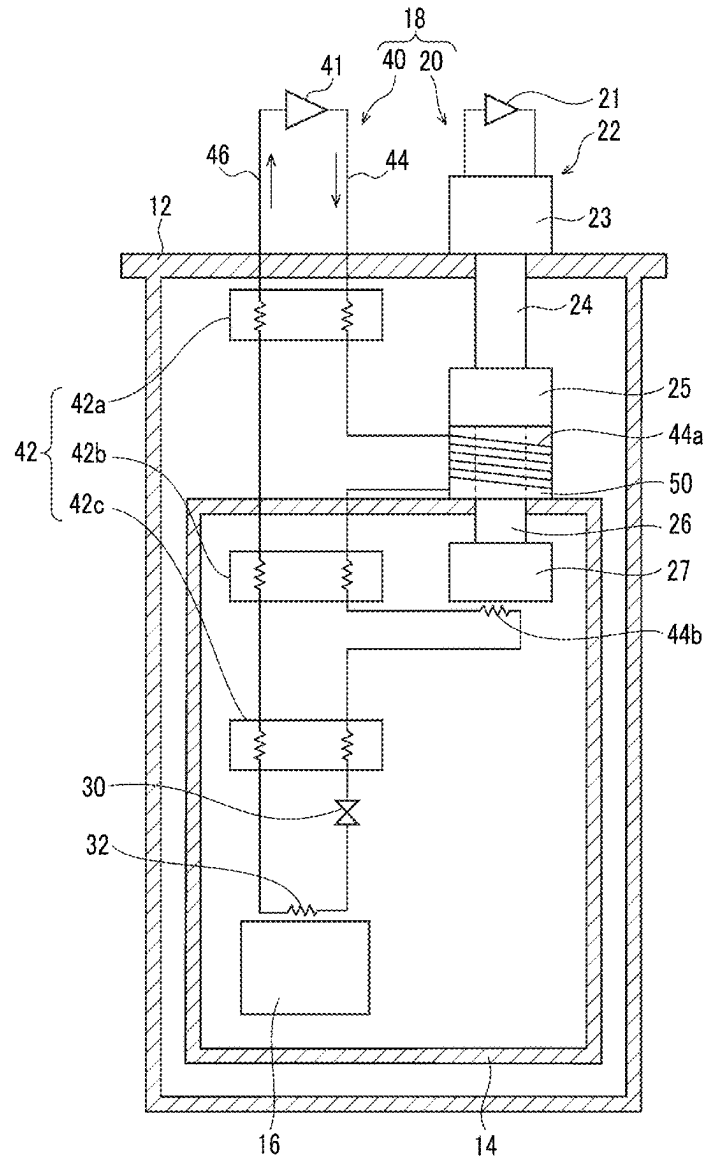
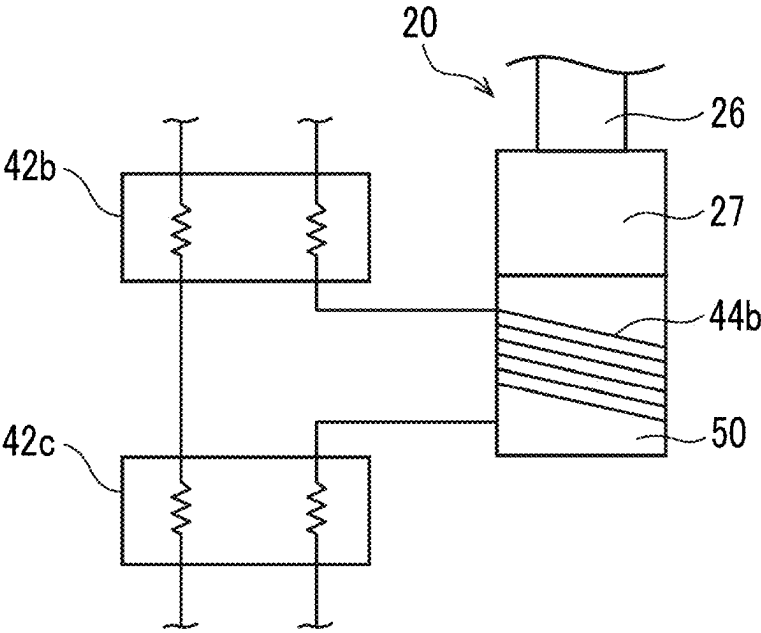


FIG. 2



JOULE-THOMSON CRYOCOOLER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a bypass continuation of International PCT Application No. PCT/JP2023/038530, filed on Oct. 25, 2023, which claims priority to Japanese Patent Application No. 2022-188372, filed on Nov. 25, 2022, which are incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

[0002] A certain embodiment of the present invention relates to a Joule-Thomson (JT) cryocooler.

Description of Related Art

[0003] In the related art, there has been known a JT cryocooler that includes a JT valve that enables cooling of a refrigerant gas using JT expansion, and a pre-cooling cryocooler such as a Gifford-McMahon (GM) cryocooler that pre-cools the refrigerant gas supplied to the JT valve.

SUMMARY

[0004] According to an embodiment of the present invention, there is provided a Joule-Thomson cryocooler including a pre-cooling cryocooler that includes a pre-cooling stage, a stage extension component that is attached to the pre-cooling stage and is cooled by the pre-cooling stage, and a refrigerant pipe that is mounted on the stage extension component to enable heat exchange with the stage extension component.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram schematically showing a cryogenic cooling device according to an embodiment.

[0006] FIG. 2 is a diagram schematically showing another example of a stage extension component according to the embodiment.

DETAILED DESCRIPTION

[0007] In general, a refrigerant pipe of a JT cryocooler is mounted by being directly wound around a cooling stage of a pre-cooling cryocooler. A refrigerant gas flowing through the refrigerant pipe is cooled to a target pre-cooling temperature by heat exchange with the cooling stage. However, in a case where the cooling stage is small in size and has a small surface area, a heat exchange area on which the refrigerant pipe is mounted may be insufficient, and the pre-cooling of the refrigerant gas may be insufficient.

[0008] It is desirable to expand the heat exchange area of the pre-cooling cryocooler of the JT cryocooler.

[0009] Hereinafter, an embodiment for carrying out the present invention will be described in detail with reference to the drawings. In the description and the drawings, the same or equivalent components, members, and processes are denoted by the same reference numerals, and overlapping description is omitted as appropriate. The scale and the shape of each of parts illustrated in the drawings are set for convenience to make the description easy to understand, and are not to be interpreted as limiting unless stated otherwise. The embodiment is merely an example and does not limit the

scope of the present invention. All features described in the embodiment or combinations thereof are not necessarily essential to the present invention.

[0010] FIG. 1 is a diagram schematically showing a cryogenic cooling device 10 according to an embodiment. The cryogenic cooling device 10 includes a vacuum chamber 12, a radiation shield 14, and a JT cryocooler 18 for cooling an object 16 to be cooled.

[0011] The vacuum chamber 12 may be, for example, a cryostat, and is configured to provide a cryogenic vacuum environment therein. The vacuum chamber 12 is formed of a metal material such as stainless steel or other suitable high-strength material to withstand ambient pressure (for example, atmospheric pressure). The radiation shield 14, a low-temperature section of the JT cryocooler 18, and the object 16 to be cooled are disposed in the vacuum chamber 12.

[0012] The radiation shield 14 is disposed in the vacuum chamber 12 to surround the low-temperature section of the JT cryocooler 18 and the object 16 to be cooled, and suppresses inflow of radiant heat from the vacuum chamber 12 to the JT cryocooler 18 and the object 16 to be cooled. The radiation shield 14 is formed of a high thermal conductivity metal material such as copper (for example, pure copper). An insulator such as a multi-layer insulator may be disposed between the vacuum chamber 12 and the radiation shield 14.

[0013] The object to be cooled 16 may be, for example, a superconducting device such as a superconducting coil, a measuring device that operates better at a cryogenic temperature, or other devices for use at a cryogenic temperature. Alternatively, the object 16 to be cooled may be, for example, a cryogenic fluid such as liquid helium, and the JT cryocooler 18 may be used for recondensing a vaporized cryogenic fluid.

[0014] The JT cryocooler 18 includes a pre-cooling cryocooler 20 and a refrigerant circuit 40 including a JT valve 30 and a final heat exchanger 32. A refrigerant flowing through the refrigerant circuit 40 is pre-cooled by the pre-cooling cryocooler 20, further cooled by JT expansion in the JT valve 30, and is supplied to the final heat exchanger 32. The object 16 to be cooled is cooled by heat exchange with the final heat exchanger 32. The cooled refrigerant is collected from the final heat exchanger 32, pressurized by a compressor described below, pre-cooled by the pre-cooling cryocooler 20 again, and is supplied to the JT valve 30. In this way, the refrigerant circulates through the refrigerant circuit 40. The JT cryocooler 18 is capable of cooling the final heat exchanger 32 to a temperature range of, for example, about 4 K or lower (for example, 1 K to 4 K), and thus, the object 16 to be cooled can be cooled to the temperature range.

[0015] The pre-cooling cryocooler 20 is, for example, a two-stage GM cryocooler. The pre-cooling cryocooler 20 includes a first compressor 21 and an expander 22 also referred to as a cold head. The expander 22 includes a drive unit 23, a first cylinder 24, a first pre-cooling stage 25, a second cylinder 26, and a second pre-cooling stage 27. The first compressor 21 is disposed in an ambient environment (for example, room temperature and atmospheric pressure environment), that is, outside the vacuum chamber 12. The expander 22 is installed in the vacuum chamber 12 such that the drive unit 23 is disposed outside the vacuum chamber 12 and the cylinders and the pre-cooling stages are disposed inside the vacuum chamber 12.

[0016] The first cylinder 24 connects the first pre-cooling stage 25 to the drive unit 23, so that the first pre-cooling stage 25 is structurally supported by the drive unit 23. The second cylinder 26 connects the second pre-cooling stage 27 to the first pre-cooling stage 25, so that the second pre-cooling stage 27 is structurally supported by the first pre-cooling stage 25. The first cylinder 24 and the second cylinder 26 extend coaxially, and the drive unit 23, the first cylinder 24, the first pre-cooling stage 25, the second cylinder 26, and the second pre-cooling stage 27 are linearly arranged in a line in this order. Typically, the first pre-cooling stage 25 and the second pre-cooling stage 27 are formed of a high thermal conductivity metal material such as copper (for example, pure copper), and the first cylinder 24 and the second cylinder 26 are formed of other metal materials such as stainless steel.

[0017] A first displacer and a second displacer (not shown) are disposed reciprocally in an inside of each of the first cylinder 24 and the second cylinder 26. A first regenerator and a second regenerator (not shown) are incorporated into the first displacer and the second displacer, respectively. In addition, the drive unit 23 includes a drive mechanism (not shown) such as a motor for reciprocating the first displacer and the second displacer. The drive mechanism includes a flow path switching mechanism that switches a flow path of a refrigerant gas to periodically repeat supply and exhaust of the refrigerant gas to and from an inside of the expander 22. A refrigerant gas of the pre-cooling cryocooler 20 is usually a helium gas, but other appropriate gases may be used.

[0018] The first compressor 21 is configured to collect the refrigerant gas from the expander 22, to pressurize the collected refrigerant gas, and to supply the refrigerant gas to the expander 22 again. By circulating the refrigerant gas between the first compressor 21 and the expander 22 with an appropriate combination of a pressure fluctuation and a volume fluctuation of the refrigerant gas in the expander 22, a thermodynamic cycle (for example, a GM cycle) for cold generation is configured, and the expander 22 can provide cryogenic cooling.

[0019] The first pre-cooling stage 25 is cooled to a first cooling temperature, and the second pre-cooling stage 27 is cooled to a second cooling temperature lower than the first cooling temperature. The first cooling temperature may be selected from a temperature range of, for example, 50 K or higher and 150 K or lower. The second cooling temperature may be selected from a temperature range of, for example, 10 K or higher and 25 K or lower.

[0020] In this embodiment, a stage extension component 50 is attached to the first pre-cooling stage 25. The stage extension component 50 is thermally coupled to the first pre-cooling stage 25 and is cooled to the first cooling temperature by the first pre-cooling stage 25. Therefore, the stage extension component 50 can be integrated with the first pre-cooling stage 25 and can function as a part of the first pre-cooling stage 25.

[0021] The stage extension component 50 is formed of, for example, a metal material having a high thermal conductivity, such as copper (for example, pure copper or a copper alloy) or aluminum (for example, pure aluminum or an aluminum alloy), or other high thermal conductivity material. Here, the high thermal conductivity material may be, for example, a material having a higher thermal conductivity than stainless steel (for example, SUS304).

[0022] The stage extension component 50 has a tubular (for example, cylindrical) shape, is fixed to the first pre-cooling stage 25 at one end, and extends toward the second pre-cooling stage 27. The second cylinder 26 is disposed in the stage extension component 50. A length in an axial direction of the stage extension component 50 is shorter than that of the second cylinder 26. Therefore, the other end of the stage extension component 50 does not reach the second pre-cooling stage 27. Therefore, the stage extension component 50 is not in physical contact with the second pre-cooling stage 27.

[0023] The stage extension component 50 is detachably attached to the first pre-cooling stage 25. For example, a flange formed at one end of the stage extension component 50 may be fixed to the first pre-cooling stage 25 by a fastening member such as a bolt. By detaching the fastening member, the stage extension component 50 can be released from the first pre-cooling stage 25. The stage extension component 50 may be attached to the first pre-cooling stage 25 by a non-detachable method such as brazing or welding.

[0024] As shown in FIG. 1, the stage extension component 50 may connect the first pre-cooling stage 25 to the radiation shield 14. For example, a flange formed at an end portion of the stage extension component 50 on a side opposite to the first pre-cooling stage 25 may be fixed to the radiation shield 14 by a fastening member such as a bolt. The radiation shield 14 may be thermally coupled to the first pre-cooling stage 25 via the stage extension component 50 and may be cooled to the first cooling temperature by the first pre-cooling stage 25. The radiation shield 14 surrounds the low-temperature section of the JT cryocooler 18, such as the second pre-cooling stage 27, and can suppress heat inflow to the low-temperature section.

[0025] The radiation shield 14 may be directly attached to the first pre-cooling stage 25 without using the stage extension component 50, or may be thermally coupled to the first pre-cooling stage 25 via a heat transfer member separate from the stage extension component 50.

[0026] The refrigerant circuit 40 includes a second compressor 41, a heat exchanger group 42, and a refrigerant supply line 44 and a refrigerant collection line 46 that connect these components in addition to the JT valve 30 and the final heat exchanger 32. A refrigerant gas that circulates through the refrigerant circuit 40 is usually a helium gas, but other appropriate gases may be used. The refrigerant circuit 40 is not limited to the specific configuration described herein and can adopt various typical configurations as appropriate.

[0027] The second compressor 41 is configured to pressurize the refrigerant gas collected from the refrigerant collection line 46 and to send the collected refrigerant gas to the refrigerant supply line 44. For understanding, a direction in which the refrigerant flows is indicated by an arrow in FIG. 1. The second compressor 41 serves as a refrigerant source that circulates the refrigerant in the refrigerant circuit 40. The second compressor 41 is disposed outside the vacuum chamber 12.

[0028] The heat exchanger group 42 in the refrigerant circuit 40 is disposed between the second compressor 41 and the final heat exchanger 32. The heat exchanger group 42 is composed of a series of counterflow heat exchangers (42a to 42c), and has a three-stage configuration of a first heat exchanger 42a, a second heat exchanger 42b, and a third heat exchanger 42c in the present embodiment. The first heat

exchanger 42a is disposed between the vacuum chamber 12 and the radiation shield 14, that is, in a space inside the vacuum chamber 12 and outside the radiation shield 14. The second heat exchanger 42b, the third heat exchanger 42c, and the final heat exchanger 32 are disposed inside the radiation shield 14.

[0029] The first heat exchanger 42a cools a high-temperature (for example, a room temperature, for example, approximately 300 K) refrigerant gas flowing into the vacuum chamber 12 from the outside of the vacuum chamber 12. The second heat exchanger 42b further cools the refrigerant cooled by the first heat exchanger 42a and the first pre-cooling stage 25. The third heat exchanger 42c further cools the refrigerant cooled by the second heat exchanger 42b and the second pre-cooling stage 27.

[0030] The refrigerant supply line 44 connects a discharge side of the second compressor 41 to a refrigerant inlet of the final heat exchanger 32, and the refrigerant collection line 46 connects a refrigerant outlet of the final heat exchanger 32 to a suction side of the second compressor 41. The refrigerant supply line 44 includes a high pressure side flow path of each of the first heat exchanger 42a, the second heat exchanger 42b, and the third heat exchanger 42c, and the refrigerant collection line 46 includes a low pressure side flow path of each of the first heat exchanger 42a, the second heat exchanger 42b, and the third heat exchanger 42c. The refrigerant flowing through the high pressure side flow path can be cooled by heat exchange between the high pressure side flow path and the low pressure side flow path in each heat exchanger. The high pressure side flow path and the low pressure side flow path may be called a high temperature side flow path and a low temperature side flow path, respectively.

[0031] In addition, the refrigerant supply line 44 includes a first refrigerant pipe 44a and a second refrigerant pipe 44b. The refrigerant pipes are formed of, for example, a high thermal conductivity metal material such as copper (for example, pure copper).

[0032] The first refrigerant pipe 44a extends from the first heat exchanger 42a via the first pre-cooling stage 25 to the second heat exchanger 42b. The first refrigerant pipe 44a connects the high pressure side flow path of the first heat exchanger 42a to the high pressure side flow path of the second heat exchanger 42b. The first refrigerant pipe 44a is thermally coupled to the first pre-cooling stage 25, and the refrigerant flowing through the first refrigerant pipe 44a is cooled by the first pre-cooling stage 25.

[0033] Note that, in this embodiment, the first refrigerant pipe 44a is mounted on the stage extension component 50 to enable heat exchange with the stage extension component 50, instead of the first pre-cooling stage 25. For example, the first refrigerant pipe 44a is fixed to the stage extension component 50 in a state of being wound around an outer peripheral surface of the stage extension component 50. The first refrigerant pipe 44a is not wound around the first pre-cooling stage 25. As described above, the stage extension component 50 is attached and thermally coupled to the first pre-cooling stage 25, so that the first refrigerant pipe 44a is cooled to the first cooling temperature by the stage extension component 50 cooled by the first pre-cooling stage 25.

[0034] The second refrigerant pipe 44b extends from the second heat exchanger 42b via the second pre-cooling stage 27 to the third heat exchanger 42c. The second refrigerant

pipe 44b connects the high pressure side flow path of the second heat exchanger 42b to the high pressure side flow path of the third heat exchanger 42c. The second refrigerant pipe 44b is thermally coupled to the second pre-cooling stage 27, and the refrigerant flowing through the second refrigerant pipe 44b is cooled by the second pre-cooling stage 27. The second refrigerant pipe 44b may be fixed to the second pre-cooling stage 27 in a state of being wound around an outer peripheral surface of the second pre-cooling stage 27.

[0035] The JT valve 30 is disposed between the last heat exchanger of the heat exchanger group 42 (in the present example, the third heat exchanger 42c) and the final heat exchanger 32 in the refrigerant supply line 44. The high pressure side flow path of the third heat exchanger 42c is connected to the refrigerant inlet of the final heat exchanger 32 via the JT valve 30. The JT valve 30 is a fixed orifice in the present embodiment. However, the JT valve 30 may be a variable orifice of which an opening degree is adjustable.

[0036] In the steady operation of the JT cryocooler 18, the refrigerant flows through the refrigerant circuit 40 as follows. The high-pressure refrigerant compressed by the second compressor 41 is first supplied to the high pressure side flow path of the first heat exchanger 42a. The high-pressure refrigerant flowing through the high pressure side flow path of the first heat exchanger 42a is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the first heat exchanger 42a. The high-pressure refrigerant cooled by the first heat exchanger 42a flows into the first refrigerant pipe 44a.

[0037] The high-pressure refrigerant flowing through the first refrigerant pipe 44a is cooled by the first pre-cooling stage 25 of the pre-cooling cryocooler 20 and is sent into the high pressure side flow path of the second heat exchanger 42b. The high-pressure refrigerant flowing through the high pressure side flow path of the second heat exchanger 42b is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the second heat exchanger 42b. The high-pressure refrigerant cooled by the second heat exchanger 42b flows into the second refrigerant pipe 44b.

[0038] The high-pressure refrigerant flowing through the second refrigerant pipe 44b is cooled by the second pre-cooling stage 27 of the pre-cooling cryocooler 20 and is sent into the high pressure side flow path of the third heat exchanger 42c. The high-pressure refrigerant flowing through the high pressure side flow path of the third heat exchanger 42c is cooled by heat exchange with the returning low-pressure refrigerant flowing through the low pressure side flow path of the third heat exchanger 42c. In this way, the high-pressure refrigerant is cooled to a temperature (that is, a temperature equal to or lower than an inversion temperature) at which the JT effect is expected, and is sent to the JT valve 30.

[0039] When the cooled high-pressure refrigerant passes through the JT valve 30, the cooled high-pressure refrigerant becomes a low-pressure refrigerant in a mist-like gas-liquid mixed state due to the Joule-Thomson effect, and generates a cooling capacity in a temperature range of the liquefied refrigerant. The mist-like low-pressure refrigerant is sent to the final heat exchanger 32. As described above, in a case where the refrigerant is helium, the final heat exchanger 32 can be cooled to a liquid helium temperature range. The final

heat exchanger 32 can cool the object 16 to be cooled to the temperature through heat exchange with the object 16 to be cooled.

[0040] When cooling the final heat exchanger 32, the mist-like low-pressure refrigerant evaporates and vaporizes again. The unliquefied refrigerant and the refrigerant that is vaporized by evaporation in the JT valve 30 are returned to the low pressure side flow path of the third heat exchanger 42c. The low-pressure refrigerant flows through the refrigerant collection line 46 in the order of the third heat exchanger 42c, the second heat exchanger 42b, and the first heat exchanger 42a. In this case, the low-pressure refrigerant is heated while cooling the high-pressure refrigerant in each of the heat exchangers (42c, 42b, and 42a) as described above. In this way, the low-pressure refrigerant that has returned to the room temperature exits the vacuum chamber 12, is collected in the second compressor 41, and is compressed again.

[0041] In this way, the cryogenic cooling device 10 can cool the object 16 to be cooled to a desired temperature lower than the second cooling temperature of the pre-cooling cryocooler 20, for example, about 4 K or lower (for example, 1 K to 4 K).

[0042] According to the embodiment, the stage extension component 50 is attached to the first pre-cooling stage 25, thereby expanding a heat exchange area of one stage of the pre-cooling cryocooler 20. By adjusting the shape of the stage extension component 50, it is possible to secure an area required to sufficiently pre-cool the first refrigerant pipe 44a. For example, a surface area of the stage extension component 50 can be increased by increasing dimensions of the stage extension component 50, such as by increasing its length in the axial direction or increasing its diameter. Even when the first pre-cooling stage 25 is small in size and has a small surface area, the stage extension component 50 can compensate for the lack of the heat exchange area for the first refrigerant pipe 44a.

[0043] As a result, the pre-cooling cryocooler 20 can adopt a general-purpose product. For example, a commercially available GM cryocooler can be used as the pre-cooling cryocooler 20. It is no longer necessary to specially design the surface area of the first pre-cooling stage 25 for pre-cooling the JT cryocooler 18, which leads to reduction in the cost of the JT cryocooler 18.

[0044] In addition, in this embodiment, the stage extension component 50 is detachably attached to the first pre-cooling stage 25. The first refrigerant pipe 44a is not mounted on the first pre-cooling stage 25. Therefore, the first pre-cooling stage 25 can be separated from the stage extension component 50 by releasing the stage extension component 50 from the first pre-cooling stage 25. This is convenient, for example, as it facilitates the detachment of the pre-cooling cryocooler 20 from the JT cryocooler 18 for maintenance of the pre-cooling cryocooler 20.

[0045] The present invention has been described above based on the examples. It will be understood by those skilled in the art that the present invention is not limited to the above-described embodiment, various design changes are possible, various modification examples are possible, and such modification examples are also within the scope of the present invention. Various characteristics described in relation to one embodiment are also applicable to other embodiments. A new embodiment generated through combination also has the effects of each of the combined embodiments.

[0046] FIG. 2 is a diagram schematically showing another example of the stage extension component 50 according to the embodiment. As shown in the figure, the stage extension component 50 may be attached to the second pre-cooling stage 27 of the pre-cooling cryocooler 20. The stage extension component 50 may be thermally coupled to the second pre-cooling stage 27 and may be cooled to the second cooling temperature by the second pre-cooling stage 27. The stage extension component 50 may be integrated with the second pre-cooling stage 27 and may function as a part of the second pre-cooling stage 27. The stage extension component 50 may have a tubular (for example, cylindrical) shape, be fixed to the second pre-cooling stage 27 at one end, and extend in the axial direction from the second pre-cooling stage 27 to a side opposite to the second cylinder 26.

[0047] The second refrigerant pipe 44b may be mounted on the stage extension component 50 to enable heat exchange with the stage extension component 50. The second refrigerant pipe 44b connects the second heat exchanger 42b to the third heat exchanger 42c as described above. The second refrigerant pipe 44b is fixed to the stage extension component 50 in a state of being wound around an outer peripheral surface of the stage extension component 50. In this example, the second refrigerant pipe 44b is not wound around the second pre-cooling stage 27. The stage extension component 50 is attached and thermally coupled to the second pre-cooling stage 27, so that the second refrigerant pipe 44b is cooled to the second cooling temperature by the stage extension component 50 cooled by the second pre-cooling stage 27.

[0048] In this way, the heat exchange area of the two stages of the pre-cooling cryocooler 20 can be expanded by using the stage extension component 50. By adjusting the shape of the stage extension component 50, it is possible to secure an area required to sufficiently pre-cool the second refrigerant pipe 44b. Even when the second pre-cooling stage 27 is small in size and has a small surface area, the stage extension component 50 can compensate for the lack of the heat exchange area for the second refrigerant pipe 44b.

[0049] As in the above-described embodiment, the stage extension component 50 may be detachably attached to the second pre-cooling stage 27. The second refrigerant pipe 44b is not mounted on the second pre-cooling stage 27. Therefore, the second pre-cooling stage 27 can be separated from the stage extension component 50 by releasing the stage extension component 50 from the second pre-cooling stage 27. This is convenient, for example, as it facilitates the detachment of the pre-cooling cryocooler 20 from the JT cryocooler 18 for maintenance of the pre-cooling cryocooler 20.

[0050] The stage extension component 50 shown in FIG. 2 may be combined with the embodiment described with reference to FIG. 1. That is, the JT cryocooler 18 may include a first stage extension component 50 attached to the first pre-cooling stage 25 and cooled by the first pre-cooling stage 25, and a second stage extension component 50 separate from the first stage extension component 50, attached to the second pre-cooling stage 27, and cooled by the second pre-cooling stage 27.

[0051] If necessary, for example, in a case where it is desired to further increase the heat exchange area, the first refrigerant pipe 44a may be mounted on the first pre-cooling stage 25 in addition to the stage extension component 50.

Similarly, the second refrigerant pipe **44b** may be mounted on the second pre-cooling stage **27** in addition to the stage extension component **50**.

[0052] In the embodiment of FIG. 1, the stage extension component **50** extends from the first pre-cooling stage **25** toward the second pre-cooling stage **27**, but the disposition of the stage extension component **50** is not limited thereto. The stage extension component **50** may extend from the first pre-cooling stage **25** along the first cylinder **24**. In this case, the first cylinder **24** is disposed inside the stage extension component **50**. The length in the axial direction of the stage extension component **50** may be shorter than that of the first cylinder **24** so that the stage extension component **50** is not in physical contact with the vacuum chamber **12**.

[0053] Similarly, in the embodiment of FIG. 2, the stage extension component **50** may extend from the second pre-cooling stage **27** toward the first pre-cooling stage **25** along the second cylinder **26**. The second cylinder **26** is disposed inside the stage extension component **50**. The length in the axial direction of the stage extension component **50** may be shorter than that of the second cylinder **26** so that the stage extension component **50** is not in physical contact with the first pre-cooling stage **25**.

[0054] The pre-cooling cryocooler **20** is not limited to the GM cryocooler. The pre-cooling cryocooler **20** may be a cryocooler of another type such as a pulse tube cryocooler or a Stirling cryocooler.

[0055] Although the present invention has been described using specific words and phrases based on the embodiment, the embodiment merely shows one aspect of the principle and application of the present invention, and various modifications and improvements can be made within the scope of the present invention described in claims.

[0056] It should be understood that the invention is not limited to the above-described embodiment, but may be modified into various forms on the basis of the spirit of the invention. Additionally, the modifications are included in the scope of the invention.

What is claimed is:

1. A Joule-Thomson cryocooler comprising:

- a pre-cooling cryocooler that comprises a pre-cooling stage;
- a stage extension component that is attached to the pre-cooling stage and is cooled by the pre-cooling stage; and

a refrigerant pipe that is mounted on the stage extension component to enable heat exchange with the stage extension component.

2. The Joule-Thomson cryocooler according to claim 1, wherein the stage extension component is detachably attached to the pre-cooling stage.

3. The Joule-Thomson cryocooler according to claim 2, wherein the refrigerant pipe is not mounted on the pre-cooling stage.

4. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a cylinder extending from the pre-cooling stage, and the cylinder is disposed inside the stage extension component.

5. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a cylinder extending from the pre-cooling stage, and the stage extension component extends from the pre-cooling stage to a side opposite to the cylinder.

6. The Joule-Thomson cryocooler according to claim 1, wherein the pre-cooling cryocooler comprises a first pre-cooling stage and a second pre-cooling stage that is cooled to a lower temperature than the first pre-cooling stage, and

the stage extension component is attached to the first pre-cooling stage and is cooled by the first pre-cooling stage.

7. The Joule-Thomson cryocooler according to claim 6, wherein the stage extension component is detachably attached to the first pre-cooling stage.

8. The Joule-Thomson cryocooler according to claim 7, wherein the refrigerant pipe is not mounted on the first pre-cooling stage.

9. The Joule-Thomson cryocooler according to claim 6, wherein the pre-cooling cryocooler comprises a cylinder that connects the second pre-cooling stage to the first pre-cooling stage, and

the cylinder is disposed inside the stage extension component.

10. The Joule-Thomson cryocooler according to claim 6, wherein the stage extension component is not in physical contact with the second pre-cooling stage.

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